LAB #2

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Task 1

Ipconfig prompt output in my Computer

```
Connection-specific DNS Suffix  . :
IPv6 Address. . . . . . . . . . . : 2001:4451:85db:3100:d38d:9242:ea7a:184d
Temporary IPv6 Address. . . . . . : 2001:4451:85db:3100:a8b3:8ce3:54db:358b
Link-local IPv6 Address . . . . . : fe80::e8af:1e68:9e4e:d5a6%14
IPv4 Address. . . . . . . . . . . : 192.168.1.3
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . : fe80::1%14
                                    192.168.1.1
```

Recorded tcp/ip information in my Computer

IP address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Questions

Compare the TCP/IP configuration of this computer to others on the LAN

Some similarities I noticed are the similar first 3 octets in the Ipv4 address, same subnet mask, and same default gateway.

What is similar about the IP addresses?

The first 3 octets are similar to both devices.

What is similar about the default gateways?

They are exactly the same.

The IP addresses should share the same network portion. All machines in the LAN should share the same default gateway. Record a couple of the IP Addresses:

192.168.1.3

192.168.1.4

192.168.1.6

192.168.1.13

Task 2

Results

ping another computer’s IP Adress from the previous lab.

Successful

```
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=94ms TTL=64
Reply from 192.168.1.13: bytes=32 time=12ms TTL=64
Reply from 192.168.1.13: bytes=32 time=124ms TTL=64
Reply from 192.168.1.13: bytes=32 time=49ms TTL=64
```

ping the IP address of the default gateway

Successful

```
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
```

ping the IP address of a DHCP or DNS servers

Successful

```
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
```

ping the Loopback IP address of this computer

Successful

```
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

Ping the Cisco website

Successful

```
C:\Users\Migs>ping www.cisco.com

Pinging e2867.dsca.akamaiedge.net [2001:4450:39:191::b33] with 32 bytes of data:
Reply from 2001:4450:39:191::b33: time=3ms
Reply from 2001:4450:39:191::b33: time=2ms
Reply from 2001:4450:39:191::b33: time=2ms
Reply from 2001:4450:39:191::b33: time=2ms

Ping statistics for 2001:4450:39:191::b33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

Successful

```
C:\Users\Migs>ping www.microsoft.com

Pinging e13678.dscb.akamaiedge.net [2001:4450:39:19d::356e] with 32 bytes of data
Reply from 2001:4450:39:19d::356e: time=3ms
Reply from 2001:4450:39:19d::356e: time=2ms
Reply from 2001:4450:39:19d::356e: time=3ms
Reply from 2001:4450:39:19d::356e: time=3ms

Ping statistics for 2001:4450:39:19d::356e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\Migs>ping www.msn.de
Ping request could not find host www.msn.de. Please check the name and try again.

C:\Users\Migs>ping msn.de

Pinging msn.de [131.253.33.203] with 32 bytes of data:
Reply from 131.253.33.203: bytes=32 time=34ms TTL=116
Reply from 131.253.33.203: bytes=32 time=36ms TTL=116
Reply from 131.253.33.203: bytes=32 time=35ms TTL=116
Reply from 131.253.33.203: bytes=32 time=35ms TTL=116

Ping statistics for 131.253.33.203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

Trace the route to the Cisco web site

Successful

```
C:\Users\Migs>tracert www.cisco.com

Tracing route to e2867.dsca.akamaiedge.net [2001:4450:39:1a9::b33]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  2001:4451:8084::1:301a
  2     4 ms     3 ms     2 ms  2001:4450:10:1e2::1
  3     3 ms     3 ms     2 ms  2001:4450:10:1e3::
  4     3 ms     2 ms     2 ms  2001:4450:0:6000::8e
  5     *        *        3 ms  2001:4450:0:6000::64
  6     2 ms     2 ms     2 ms  2001:4450:39:1a9::b33

Trace complete.
```

Trace other IP addresses or domain names

Successful

```
C:\Users\Migs>tracert google.com

Tracing route to google.com [2404:6800:4017:803::200e]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  2001:4451:8084::1:301a
  2     4 ms     3 ms     3 ms  2001:4450:10:1e2::1
  3     3 ms     2 ms     3 ms  2001:4450:10:1e2::
  4     2 ms     2 ms     2 ms  2001:4450:0:6000::8c
  5     2 ms     *        *     2001:4450:0:6000::9f
  6     3 ms     3 ms     3 ms  2001:4860:1:1::1916
  7     3 ms     2 ms     2 ms  2404:6800:832f::1
  8     4 ms     3 ms     3 ms  2001:4860:0:1::6cae
  9     3 ms     2 ms     2 ms  2001:4860:0:1::193c
 10     4 ms     4 ms     3 ms  2001:4860:0:1::3d4d
 11     2 ms     2 ms     2 ms  2001:4860:0:1::2f09
 12     3 ms     2 ms     3 ms  mnl08s01-in-x0e.1e100.net [2404:6800:4017:803::200e]

Trace complete.
```

Successful

```
Tracing route to yahoo.com [2001:4998:24:120d::1:0]
over a maximum of 30 hops:

  1     1 ms     1 ms    <1 ms  2001:4451:8084::1:301a
  2     4 ms     3 ms     3 ms  2001:4450:10:1e2::1
  3     3 ms     3 ms     2 ms  2001:4450:10:1e2::
  4     9 ms     2 ms     3 ms  2001:4450:0:6000::8c
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8   167 ms   166 ms   167 ms  br03.lax05.as3491.net [2400:8800:7e02::11]
  9     *        *        *     Request timed out.
 10   191 ms   189 ms   189 ms  ae-0.pat2.laz.yahoo.com [2001:4998:f02f:b::1]
 11   195 ms   189 ms   190 ms  ae8.pat2.pao.yahoo.com [2001:4998:f002:227::]
 12   207 ms   210 ms   210 ms  ae-6.pat2.swp.yahoo.com [2001:4998:f002:206::1]
 13   220 ms   218 ms   209 ms  ae-10.pat2.gqb.yahoo.com [2001:4998:f007:200::1]
 14   217 ms   209 ms   213 ms  2001:4998:f00f:207::1
 15   209 ms   203 ms   204 ms  et-0-0-0.clr1-a-gdc.gq1.yahoo.com [2001:4998:c:fe21::1]
 16   209 ms   209 ms   209 ms  et7-1.fab8-2-gdc.gq1.yahoo.com [2001:4998:c:fa7f::1]
 17   209 ms   209 ms   209 ms  et32-1.bas2-2-sta.gq1.yahoo.com [2001:4998:c:ba45::1]
 18   209 ms   210 ms   209 ms  media-router-fp73.prod.media.vip.gq1.yahoo.com [2001:4998:24:120d::1:0]
```

Trace a local host name or IP address

Successful

```
C:\Users\Migs>tracert 192.168.1.6

Tracing route to 192.168.1.6 over a maximum of 30 hops

  1    42 ms     *       22 ms  192.168.1.6

Trace complete.
```

Conclusion and Takeaways

Basic network information can be seen by the input of ipconfig, and additional information through ipconfig /all. You can verify connections by pinging other hosts thru the command ping. You can trace the route going to the host by the command tracert. I had learned a lot from the lab activity as it covered some of the basic commands you can issue for networking. Although basic commands, they are used almost everytime as an IT so learning them is very important.